

An overview of Internet protocols

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In a very short period of time, the Internet has risen from being viewed as a peculiarity of the defence and academic community of the late sixties to become a significant component of many of our working and home lives. The possible impact of this technology has the futurists talking of the collapse of national borders, the bypass of tax systems and traditional commerce and the emancipation of the individual from the clutches of government, corporations and the boundaries of culture, gender and physical constraints. Most users' experience of the Internet is through browser software and applications such as e-mail, World Wide Web, Newsgroups and media plug-ins. Beneath this glossy, graphical, multimedia world exists the architectural and protocol components which have enabled the Internet evolution to reach into so many parts of everyday life in such a short time. The aim of this paper is to outline some of these components, their characteristics and limitations, and their ongoing development to address those limitations and add new capabilities. Even though an overview of Internet protocols is included in this paper, basic knowledge of Internet technology is assumed.

1. Introduction

The Internet protocol (IP) is one of a large number of network layer protocols. Clearly the explosive growth of the Internet alone makes IP an important factor in considering business strategy and evolution. There are, however, underlying reasons why IP has gained the status that it has:

- IP is an internetworking protocol, which is designed to operate over almost all underlying networks, and there is no requirement therefore for a homogeneous end-to-end infrastructure — end stations may use a variety of cheap interface cards (e.g. Ethernet) but the core network may be comprised of vastly different technologies,
- IP is connectionless, which is well suited to short-lived communications, but can also support long-lived connections (via the transmission control protocol (TCP)),
- IP networks are highly scalable, with a uniformly accepted addressing scheme,
- IP is service-independent, in that a wide range of application types are supported by a single, modular protocol suite, and this means that new services are extremely easy to deploy, without requiring extensive changes to end station or network stacks.

For the above reasons, IP is becoming the preferred end-to-end network protocol not only for the public Internet and intranets, but also, to an increasing degree, for applications such as real-time multimedia applications. While the TCP/IP suite undoubtedly has many limitations, it is important to distinguish these from the well-publicised weaknesses of the public Internet, in particular the following:

- Internet congestion (which is often server rather than network congestion) arises largely from underdimensioning, which is not necessarily the case for all networks that use IP technology (it is the case, however, that in present IP networks, bandwidth management largely has to be performed at layers below IP),
- the insecurity of the Internet is not necessarily a characteristic of IP networks in general — it is quite possible to make IP networks highly secure,
- real-time applications can be well supported by the TCP/IP protocol suite, even though congestion may limit their support over the public Internet.

The main limitation of the IP suite is its lack of support for end-to-end network 'connections', as typified by traditional telco networks (PSTN, ATM, etc). While emerging standards aim to rectify this (see, for example, the resource reservation set-up protocol (RSVP) [1], many would question the suitability of this communication mode

for IP networks. The applicability of TCP/IP, however, is far wider than merely non-real-time traffic on the public Internet or intranets. It is emerging as the preferred end-to-end network protocol for all types of data, including those with real-time characteristics. From the above, it is clear that there are many services which are supported by IP networks. Most applications could be supported, given appropriate underlying networks and dimensioning.

2. Overview of the TCP/IP protocol suite

The TCP/IP protocol stack, as shown in Fig 1, has a number of differences (including the lack of a session layer) when compared to the open systems interconnection (OSI) stack. While the TCP/IP suite embodies well-tested principles of protocol layering, there is no formal definition of a protocol stack reference model, and as such the protocol layers are not rigidly defined.

As described above, IP is a connectionless network layer protocol designed for wide area networking. IP packets, or datagrams, are composed of a 20-byte header and a payload. The maximum size of the data area is 65 535 octets which may be expanded in later versions of the protocol such as IP version 6 (IPv6). The version currently in widespread use is IPv4.

IP is designed as an internetworking protocol and as such IP datagrams can be carried between gateways (or routers) on almost any link-layer protocol, such as Frame Relay, ATM or Ethernet. In the LAN, the dominant technology is Ethernet while in the WAN it is leased and switched circuits. A router is a specialised device using similar processors to personal computers but running specialised software programmes on dedicated hardware to provide optimised routing and packet forwarding capabilities. The TCP/IP architecture mandates that the TCP and IP layers should not be made specific to any sub-

network technology, thus ensuring interworking. The interface between the IP layer and link layer is where the characteristics of the specific subnet technology are taken into account. Since IP has its own addressing format and so does the link layer underneath, such as Ethernet addresses, a protocol, called the address resolution protocol (ARP), has been devised for most combinations of network and link-layer protocols. An IP router will build an ARP table as it learns about what hardware is attached to the networks to which it is attached. The Internet control management protocol (ICMP) provides basic information to hosts and routers of problems such as packet drops, unavailable addresses, etc.

Hosts run application programmes for Internet users such as e-mail, Newsgroup, Chat (IRC) and WWW (HTTP) servers and clients. In addition, they provide infrastructure and application services for user hosts which are generally inappropriate for deployment on routers themselves. Examples of this are mail, Newsgroup and Chat relays, domain name service (DNS), WWW proxy servers, service location, and gateways to other networks and services such as GSTN telephony and X.400 mail. Emerging applications include, among others:

- electronic commerce (banking, shopping),
- Internet telephony, video-on-demand and conferencing,
- multicast audio, video and data distribution,
- network gaming, distributed simulations.

The basic architectural principle of the Internet, known as the end-to-end principle, assumes that there are many more hosts in the Internet than routers and that the hosts are reasonably high performance computers. This assumption

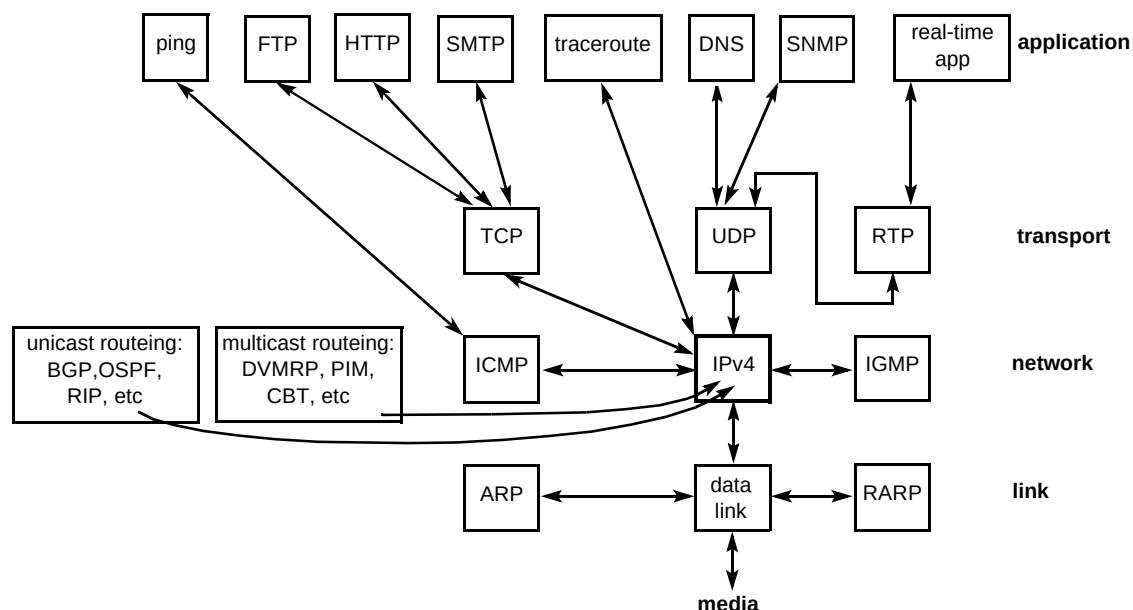


Fig 1 Outline of the TCP/IP stack.

leads to the conclusion that the routers, which have to deal with large numbers of hosts, should be as simple and generic as possible to maximise packet forwarding performance and any complexities should be pushed out into the hosts. A secondary effect of this is that the routers are simple enough to survive the rapid changes in host applications and protocols. This is often described as aiming to avoid deploying a capability on a router which could become a bottle-neck to a future application.

The main transport protocols which run over IP are the user datagram protocol (UDP), the real-time transport protocol (RTP) and the transmission control protocol (TCP). UDP is the most basic transport service and provides an unreliable, unordered, multiplexed transport service for applications. Multiplexing is achieved through the use of port numbers to differentiate between different UDP flows from the same host. Many applications have default UDP port numbers allocated by the Internet Assigned Numbers Authority (IANA). The unreliable (fire-and-forget) nature of UDP means that either the higher layer protocol must provide retransmissions (e.g. DNS, RIP (routing information protocol)) or be used by applications where retransmissions would be inappropriate, such as real-time audio and video.

It would be impossible here to give a full description of the TCP protocol [2], however, in order to appreciate the issues, the following TCP features are highlighted:

- TCP uses sliding window-based flow and congestion control, controlled by the flow of acknowledgement packets (with receiver window advertisement),
- TCP incorporates congestion control and avoidance features which are triggered by packet loss across the end-to-end path — of particular relevance are the slow-start and congestion avoidance phases [3], in which the TCP send window increases exponentially and linearly, respectively, with each round trip (all TCPs should exhibit these mechanisms both at start up and after a retransmission time out),
- standard TCP uses a cumulative positive acknowledgement mechanism, but one in which out-of sequence data can be retained — more recent enhancements support a selective acknowledgement mechanism,

- TCP uses adaptive timers, based on the round-trip time (RTT), to tune the retransmission time out.

In its approach to congestion control, TCP embodies several principles including:

- conservation of packets — a new packet cannot be injected until an old packet has left the network,
- self-clocking — packet transmissions are triggered by returning ACKs,
- slow start — as described above, the send window increases each round-trip time, to discover the sustainable throughput.

TCP provides a more comprehensive transport service than UDP in that it ensures ordered and reliable delivery in addition to port-based multiplexing. The TCP stack is therefore responsible for buffering received packets, acknowledging (ACK) the receipt to the sender, re-ordering any packets which may have arrived out of turn and reducing the sending rate in response to packet loss (congestion control) to ensure fair sharing of the available resources with other TCP connections.

The pipeline model of Fig 2 is often useful in visualising TCP flow characteristics. Data packets are transmitted end-to-end through data pipes with a variety of bandwidths, and buffering will often occur at the point at which the data pipe 'narrows'. Acknowledgements flow back along the return path, and in general 'clock out' more data.

While TCP is sufficient for traditional data applications, the requirements of real-time applications are very different and have resulted in the design of RTP. This is a framing protocol which uses the basic transport service of UDP but adds the required sequence numbers and timestamps to ensure ordered delivery and timely play-back of real-time voice, video and data media streams. In addition, RTP provides information as to the precise type of media carried and its encoding. RTP has also been designed to support transcoders, mixers and re-timing elements to enable more centrally managed streaming and conferencing services. An RTP stream is a unidirectional flow of RTP packets between a sender and either a single or multiple receivers. Multiple receivers can be supported by transmitting the RTP stream over a multicast or broadcast service as

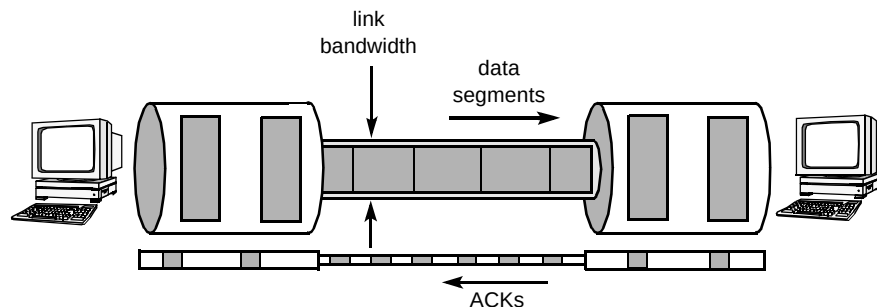


Fig 2 Pipeline model of TCP flow control.

provided by IP multicast routing. An associated control protocol has also been developed, real-time control protocol (RTCP), which provides a feedback channel from receivers to sender, between receivers, and from sender to receivers. This feedback channel provides the identifiers of the users listening to the RTP stream, information on reception quality (lost, mis-ordered, delayed packets) and other performance metrics. This information can be used to tune the sender rate to the receivers which could form the basis for a rough form of congestion control for real-time media.

3. IP over lower layers

There are a number of issues and requirements which must be addressed by any technology carrying IP. The selection of appropriate data link-layer protocols which are well suited to the transport of IP as well as to the characteristics of the underlying media is a crucial issue. There are a number of candidate lower layer protocol stacks which could be used, including:

- IP over Ethernet-like variable frames,
- IP over Frame Relay,
- IP over SMDS,
- IP over SDH, PDH or other TDM scheme,
- IP over ATM,
- IP over MPEG transport.

The criteria for the selection of the most appropriate lower layer stack comes from the IP requirements on the lower layers and some of the major requirements are outlined in the sections below.

3.1 Link management and performance

The lower layer protocol stack must maintain the link by appropriate means, and maintain a sufficient level of performance (in particular, data integrity). It is largely inappropriate to use link-layer retransmission schemes for support of IP, since, as mentioned in earlier sections, an increasing number of delay-sensitive applications use IP networking. TCP and other reliable protocols provide end-to-end reliability and flow control where necessary, but the performance generally degrades badly if the underlying links do not have good error performance.

3.2 Bandwidth management

For links within a core IP network (e.g. connecting backbone routers or remote points of presence (PoPs) to the backbone), the lower layers must simply provide dedicated bandwidth, or at least bandwidth under immediate control of the IP routers at either end of the link. Multiplexing of

different end-to-end service types (e.g. video, data, etc) is a contentious issue. It is argued by some that switched lower layers are the appropriate ones to perform service multiplexing. However, from an IP perspective, service multiplexing is performed at the IP layer (i.e. IP transporting real-time and non-real-time streams, etc), with appropriate assistance from underlying layers (see section 3.3 on support for integrated services).

Some form of bandwidth-on-demand or adaptive bandwidth allocation (e.g. the underlying network intelligently increasing or decreasing the allocated bandwidth depending on load) may be appropriate, but the implications should be considered carefully. The main difficulty arises from the fact that, increasingly, IP routers are performing intelligent packet scheduling sometimes in accordance with quality-of-service demands. Varying the bandwidth available to the router will often interfere with its ability to control packet-forwarding delays. In general, any variation in allocated bandwidth should be on a time-scale much slower than the average lifetime of an IP flow and the variation must be visible to the IP router.

There has been considerable work investigating bandwidth-on-demand mechanisms in association with particular IP flows, especially in the realm of switched virtual circuits (SVCs), e.g. ATM or FR. Some conclude that:

- applicability of this 'per flow' approach is limited, given the difficulty of recognising suitable IP flows, few of which have any associated signalling (e.g. using RSVP [1]),
- processing and memory overheads are large (e.g. per flow state in routers),
- such flows can often be supported instead by differential scheduling within the IP layer.

The following considerations should be taken into account:

- as IP flows are often dynamic and short-lived, a round-trip signalling delay per IP flow adds significant delay,
- emerging technologies will be capable of continuous presence for IP access as opposed to a pay-per-minute dial-in session,
- IP access per user often has very bursty traffic characteristics and dedicated bandwidth per user would be extremely inefficient.

3.3 Layer 2 support for IP integrated services

As described earlier, a major direction in the evolution of IP is the support for integrated services (see Braden [4])

for details). This requires scheduling and classification support in the routers and often requires some form of internetwork signalling (e.g. RSVP [1]). The end-to-end QoS requirements, which may be absolute (e.g. guaranteed delay) or relative (e.g. prioritised traffic), must be mapped down on to the component subnetworks forming the path. For high-bandwidth inter-router links, where there is often constant point-to-point bandwidth, little explicit integrated services support is needed from the link-layer technology, since QoS requirements may be satisfied purely by router scheduling. For lower bandwidth access links, or where bandwidth is shared, some QoS support is needed from the lower layers. Such QoS support must ensure that where there is contention for network resources, the differential transport is supported on demand. For IEEE 802.1p-type networks (an introduction to 802.1p can be found in Newson et al [5]), such differential transport is mainly based on a set of traffic classes. For ATM networks, such differential transport may be based on VCs with different service classes and QoS.

3.4 *Link-layer addressing and ARP support*

For multiple access with large numbers of subscribers, some form of link-layer addressing is needed to enable receivers to identify which frames are directed to them. Link-layer addressing can either be globally unique (absolute addressing), such as IEEE MAC addressing, or local addressing, such as ATM VPI/VCI addressing. The latter generally involves some form of signalling for the end points to agree upon a local address. The former needs no signalling (see section 3.5 on multicast address support). In both cases, the link layer should support address mapping between IP addresses and link-layer addresses (ARP and inverse ARP [6]).

3.5 *Support for broadcast and multicast*

Link layers should support the four forms of IP broadcast (limited, directed, subnet directed and all-subnets directed) [7]. It should also support IP multicast.

Support for IP multicast is extremely demanding in access networks which rely upon VCs. IP multicast sessions are dynamic in nature, as is IP multicast addressing. Thus, to support IP multicast over VC-based access networks, an address look-up table must be maintained and updated as receivers join and leave [8], and VCs must be set up on demand which involves signalling between the head-end and every subscribing receiver. A simpler but less efficient alternative would be to treat all multicasts as a broadcast, possibly over a permanently established broadcast VC and allow receivers to discard unwanted IP multicast traffic after decoding.

In connectionless LAN technologies, support for IP

can be algorithmically transformed into IEEE MAC addresses. A receiver subscribing to an IP multicast group G simply listens on the link layer to frames addressed to MAC address G', derived algorithmically from G. The only difficulty is that the IEEE MAC multicast address space is considerably smaller than the IP multicast (class D) address range, thus introducing some ambiguity.

3.6 *Support for multiple logical subnets*

In an access environment, bandwidth may need to be shared between multiple providers with their own subscriber populations. The lower layer protocols must be able to ensure logical and secure separation of the provider domains (usually referred to as logical IP subnets (LISs)).

3.7 *Support for variable length frames*

The link-layer protocol must be able to support variable-length network-layer frames, such as IP datagrams, efficiently. Almost all link layers are able to carry variable-length frames, but some, such as Ethernet-like protocols, do this using variable-length link-layer frames, while others, such as ATM, do this by fragmentation into fixed-size cells. The traditional argument in favour of the latter is that fixed-size cells enable QoS guarantees to be met simply, while still providing statistical multiplexing. With variable-length frames at the link layer, packet jitter can become high over low bandwidth links due to a large variation in frame sizes. Use of fixed-size cells, however, incurs extra bandwidth and processing overheads. The latter arises though additional fragmentation processing. The former arises when an IP datagram does not exactly fill an integer number of link-layer cells (e.g. 40-byte IP datagrams, which are very common, together with 8-byte LLC/SNAP header and 8-byte AAL5 trailer, total 56 bytes, would require two ATM cells with total size 106 bytes!). MPEG transport uses 144-byte cells, with even greater overhead. Where variable-length frames are used on low-rate access links, it is also possible to use suspend/resume capabilities to prevent large frames causing unacceptable delays to high-priority frames [9].

3.8 *Evolution of IP networks*

With the dramatic growth of the Internet, and corresponding commercial use of intranets, there has been intensive development of both the TCP/IP suite and the underlying technology. The strategy for the development of the infrastructure to support IP networks must take account of the direction which TCP/IP is taking. Some of the main areas of development are discussed in the following sections.

4. Internet access

Access protocols were developed to provide specific services for data access providers and users. The point-to-point protocol (PPP) provides a link control protocol (LCP) and multiple network control protocols (NCPs) as well as HDLC framing for multi-protocol transfer over serial links. The LCP is responsible for configuring, testing and maintaining the serial link over which data is to be transported. The NCP is responsible for announcing the data protocols to be used and configuration options for that network protocol over the serial link. These include the request and configuration of network addresses, address field compression and data compression. The state diagram for PPP is shown in Fig 3.

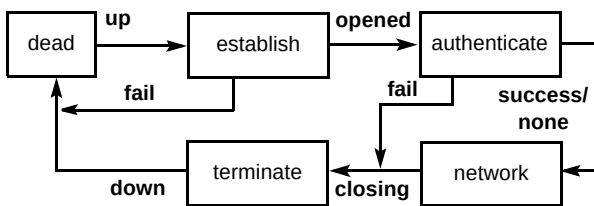


Fig 3 PPP state transition diagram.

Essentially, PPP is a negotiation protocol through which any two distant points can negotiate authentication mechanisms, protocol capabilities, and configuration parameters in order to establish a required connection. Initially it was used on PSTN lines for dial-up access, while later it was used for on-demand configuration of router-to-router back-up links especially over ISDN. Recently, PPP has been revisited by the IETF community in order to identify its capabilities as a generic mechanism to negotiate, establish and authenticate a point-to-point connection over any layer 2 technology [10].

Within the framework of PPP, the IETF is developing new techniques such as multi-link PPP (MP), bandwidth allocation protocol (BAP), integrated services over slow serial links (ISSLOW) and the layer 2 tunnelling protocol (L2TP). Multi-link PPP provides an inverse multiplexing capability in which multiple serial links are combined by the MP stack to look like a single high-speed link. MP controls the choice of which serial link a packet is sent down and can also segment a packet into several pieces to be sent in parallel down the multiple serial links [11, 12]. The BAP protocol can be used to bring up and down additional serial links within an MP bundle based on traffic demand and other performance and configuration settings [13] (see Fig 4).

The ISSLOW work is focused on supporting integrated services on narrowband links (up to 2 Mbit/s) like PSTN and ISDN, by enabling the interruption of large packets typically delivered by data applications like FTP. ISSLOW provides the controlled fragmentation of packets on a link-by-link basis so that shorter packets from lower latency

applications, such as real-time multimedia, can be inserted into the middle of the interrupted packet. This is shown diagrammatically in Fig 5. This capability should greatly enhance multimedia applications which have both data and audio components, particularly when deployed with enhanced PPP compression for TCP/IP and RTP/UDP/IP flows [9].

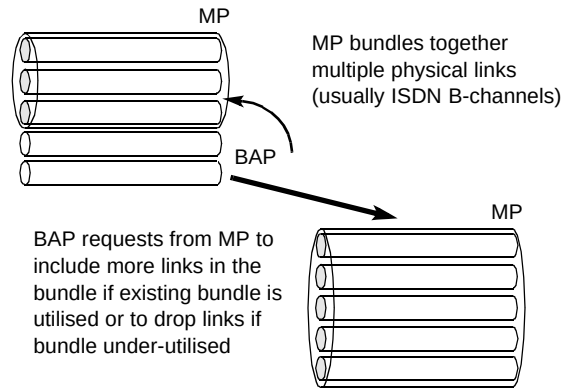


Fig 4 MP and BAP capabilities.

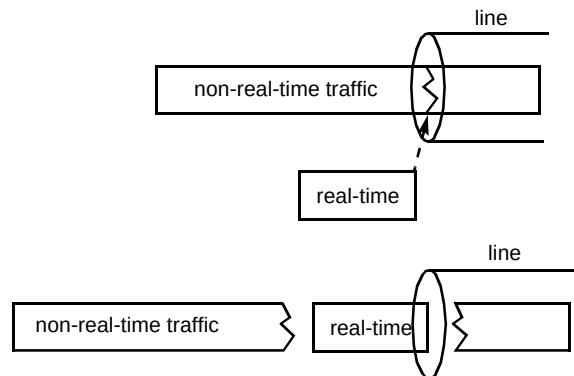


Fig 5 Packet interrupt in ISSLOW.

L2TP (layer 2 tunnelling protocol) represents an attempt to obtain a general framework Internet standard which embraces the existing proprietary capabilities provided by protocols such as Microsoft's Point to Point Tunnelling Protocol (PPTP) and Cisco's Layer 2 Forwarding (L2F). It is worth noting that this consolidation effort could be expanded to include other Internet automated tunnelling techniques such as those employed in mobile IP. L2TP conceptually modifies the PPP dial-up model so that the PPP session appears as if it terminates on a new service provider element, similar to the traditional NAS, but called the L2TP network server (LNS). However, the PPP session actually terminates at a dial-in termination element, the L2TP access concentrator (LAC), which is the source of the L2TP tunnel to the LNS. The LAC — LNS tunnel could be provided over Internet technology or layer 2 networks such as Frame Relay. This new configuration provides separation of the GSTN access concentration point from the point at which the Internet service is delivered. This enables the LAC to be provided by a network operator as a generalised

service for ISPs, which can be shared by multiple ISPs and therefore provide higher modem availability and cost savings to those ISPs.

The user's PPP session behaves as normal with the LAC undertaking a partial authentication of the user. The service address of the user is used to forward the user's IP packets into a tunnel towards the associated ISP specific LNS where service authentication and IP address allocation are undertaken (see Fig 6).

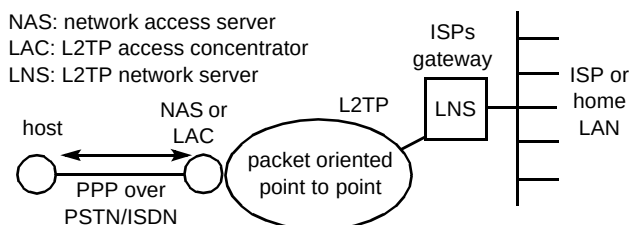


Fig 6 L2TP infrastructure.

Additional capabilities of L2TP is that it enables the host to set up an end-to-end secure tunnel to another host on the Internet and can also provide an alternative to dedicated FR SVCs by providing separation of user flows into multiple channels within a tunnel and/or multiple tunnels within a single FR PVC. In addition, because the LAC to LNS connectivity can be provided over a single large data connection rather than multiple GSTN circuits, L2TP provides statistical gain for best-effort Internet traffic. It is important to note, however, that this aggregation of traffic could compromise real-time Internet services if the underlying data network is unable to comply with integrated services and provide prioritised forwarding for specific IP real-time flows. L2TP is presently undergoing interoperability testing [14].

It is clear that Internet access is undergoing a transformation. New protocols utilise the existing access networks and new access architectures open up the way for new services. Additionally, new access technologies are coming to light like asymmetric digital subscriber line (ADSL) and cable modems which provide much more bandwidth than current technologies (up to 10 Mbps). These will need to be competitively priced if significant adoption is to occur. All the above are going to trigger more applications targeting a wider range of people and sustaining the exponential growth that the Internet enjoys today.

5. Multicast

IP multicast is designed to provide dynamic many-to-many connectivity between a set of senders and a set of receivers. The format of IP multicast datagrams is identical to that of IP unicast datagrams, being distinguished only by the fact that they carry a class D (multicast) destination IP address (in the range 224.0.0.0 to 239.255.255.255) which

denotes the multicast group. Membership of the group is dynamic and receiver-controlled — receivers inform the network that they would like to join the group. Multicast routing protocols then have the task of dynamically building a distribution tree from any sender to all subscribed receivers (for more detailed information, see the appropriate WWW pages [15]).

Unicast IP routing is generally destination based, hop-by-hop routing (or 'go to' routing). In contrast, multicast IP routing generally relies upon some form of reverse path routing (or 'come from' routing). For example, in some multicast routing protocols, the forwarder, on receipt of a multicast packet, checks whether the packet was received over the 'correct' incoming interface, that is, the one pointing back along the shortest path to the source (hence, reverse path routing). If so, the packet is forwarded on all enabled outgoing interfaces; if not, the packet is discarded. There are several candidate multicast routing protocols with widely varying characteristics. For some, the multicast distribution tree is built in a data-driven manner, for others, the tree is built by explicit control messages (e.g. join); some use a shared distribution tree (the same distribution tree used for all sources), others use a tree per source.

Through recent developments, there has been a general acceptance that multicast routing should follow the same two-level hierarchy as unicast routing, with distinct intra- and inter-domain protocols. Protocols such as distance vector multicast routing protocol (DVMRP), multicast extensions to OSPF (M-OSPF), or protocol-independent multicast (PIM) could be used for intra-domain routing. The design of the inter-domain multicast routing protocol is still under development.

A central aspect of unicast inter-domain routing (for which border gateway protocol v4, or BGP-4, is usually used) is the fact that it is policy-based routing. This enables network providers to exercise local policy decisions in the selection of paths and of transit providers. More details on BGP-4 operation are given in Whalley et al [16]. It is likely that an extension to BGP-4 will be used to provide the policy aspects of inter-domain multicast. A problem exercising the Internet community, however, is that the nature of network policies for multicast will be radically different to that for unicast. This is by virtue of the fact that unicast and multicast routing are so different, and that multicast forwarding involves a distribution tree with potentially a large degree of data replication. This is likely to result in inter-provider agreements for multicast in the Internet taking on a very different complexion from unicast.

Like IP in general, IP multicast provides an unreliable connectionless service. If reliability, ordering, flow control, access control, data naming, etc, are required, then this must be performed by the higher layers of the protocol stack. For unicast IP, reliability, flow control and ordering are

provided by the TCP protocol but it should be noted that this is only designed for one-to-one communication. Hence, to provide these functions for multicast requires new transport layer protocols, typically referred to as reliable multicast protocols, and such protocols are currently under development, with some protocols having reached deployment stage. More details on IP multicast can be found in O'Neill [17].

6. Quality of service

The Internet is evolving into a multiservice platform of the future and as such will need to support a wide range of QoS mechanisms appropriate for a diverse range of applications and commercial environments. This section describes some of the most potentially useful QoS components which may be used in building an overall QoS architecture.

6.1 Type-of-service (TOS)

In order to provide differential QoS on a packet-by-packet basis, a means must be provided by which network nodes may infer any special treatment that they should apply, using mechanisms such as those described below. For this purpose, IPv4 packet headers carry an 8-bit TOS field, which is composed of three precedence bits and five TOS bits.

The TOS bits are generally regarded as describing the type or class of service required, and this is chosen by the application. RFC 1349 defines six values with associated meanings and there are recommended values for a number of core IP applications, e.g. for Telnet the recommended value is 0×10 which signifies 'minimise delay', while for FTP data transfer the recommended value is 0×08 , which signifies 'maximise throughput'.

Use of the precedence bits is undefined, but the prevalent view is that they should be used to indicate importance (e.g. drop priority), and that their use should be under the control of the network, rather than the end system, as a tool for providing different quality of service to different users. The semantics of both fields is currently under some discussion, particularly in the light of the 'differential services' activity described below, and it is probable that some new definitions will emerge.

6.2 Priority queuing (PQ)

With priority queuing, traffic is sorted into two or possibly more queues, which represent different priorities. Scheduling is done in strict priority order, i.e. traffic is only served from a lower priority queue when the higher priority queues are empty. The TOS precedence field is one possible means of defining priorities. As with any scheme which gives one set of flows priority over another, care must be

taken to restrict the level of traffic which will be carried as high priority in order both to maintain high performance for high-priority traffic, and to avoid 'starvation' of lower priority traffic.

6.3 RED and WRED

Random-early detection (RED) is an example of a queue management, rather than a scheduling discipline, in that it controls access to, and behaviour of, a single queue, rather than multiple queues. RED monitors average queue length and compares this with two threshold values to control the rate of random packet discards, and has the overall effect of improving TCP performance. A variation on this is weighted-RED (WRED), whereby the parameters for random discard can be set according to packet classification (e.g. precedence), so giving one broad traffic class preferential throughput over another. This is a lightweight mechanism that generally imposes little overhead on router performance and is suitable for providing a small number of differential traffic classes. WRED has been applied in backbone routers, where two-level classification is provided by a single precedence bit.

6.4 Weighted fair queuing (WFQ)

In fair queuing schemes, traffic is classified into a number of flows which are placed into virtual queues. These are then serviced using a scheduling algorithm which takes precise account of packet sizes, arrival times, and current backlogs to ensure that each flow has at least the performance that would be expected if given a dedicated link with a bandwidth equal to a pre-determined proportion of the total link bandwidth. In fair queuing, each queue is allocated an equal share of the bandwidth, and in a LAN environment this can be applied 'on-the-fly', whereby routers identify currently active flows and assign each of these to a separate queue. Weighted fair queuing is a more general, and more widely useful, form whereby arbitrary shares of the bandwidth can be assigned. As well as sharing bandwidth in a controlled way, WFQ can enable precise bounds on node delay to be determined.

WFQ implementations can allow large numbers of queues, so that queues could potentially be assigned to individual user flows in order to give a very high assurance of service quality. An alternative use is in controlling the proportion of scheduling resources given to different service classes; this gives each class a guaranteed share of resources and avoids the potential for 'starvation' that is a feature of priority queuing.

Most of today's routers are heavily software-based, so that supporting complex scheduling algorithms like WFQ results in a significant performance reduction. Vendors are developing specific hardware designed to support multiple-

queue schedulers, so that such performance restrictions will be much less significant in the near future [18].

6.5 Class-based queuing (CBQ)

This is a sophisticated bandwidth-management technique, based on multi-queue scheduling, in which bandwidth allocations across a link may be assigned to different traffic categories according to a hierarchical structure. First implementations are now appearing, including at least one commercial product. A broad range of traffic classifications are possible, including by address, protocol, port number or TOS field. As an example, an important potential application could be for managing bandwidth for a set of VPNs on a common platform. In this case the top level of the hierarchy may be used to assign proportions of the total bandwidth to different corporations based on top-level addresses (see Fig 7). Lower levels of the hierarchy may then be used to further sub-divide this bandwidth, for example, first by department, then by application or protocol. In addition, for each lowest level, child-class priority levels may be set to determine relative latency performance, and 'borrowing' permissions may be granted to determine which classes are allowed to make use of any 'spare' bandwidth unused by other classes [19].

6.6 Resource reservation (RSVP)

All of the above mechanisms can be applied to provide various forms of QoS support in a connectionless environment without the need for prescriptive end-to-end standardisation and have therefore emerged with little need for consensus within the IETF. However, for the last 3 years or so, the IETF has been developing an end-to-end QoS framework based on the RSVP signalling protocol together with some service class definitions developed by the IntServ working group. This amounts to a connection-

oriented overlay which would generally co-exist with connectionless, best-effort services.

RSVP was designed for multicast applications, with the objective of supporting large numbers of receivers with differing requirements (heterogeneity). For this reason reservations are receiver-based. For a fuller description, readers are referred to White [20], but the essentials of operation are illustrated in Fig 8.

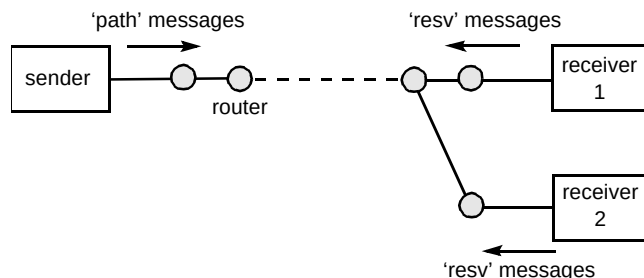


Fig 8 Resource reservation message flow.

Before any reservations are made, each sender generates *Path* messages which are sent to each receiver. These are used to carry the sender's traffic parameters, but as they pass through each router in the network path, are also used to install state and gather certain characterisation parameters. On receipt of a *Path* message, a receiver may then make a reservation by generating a *Resv* message. This propagates back along the network following the reverse path, and, if resources are available (call admission control), results in an installed reservation at all the outgoing interfaces which are downstream of the sender. The reservation is mapped to incoming data flows using a classifier (in IPv6, a flow-id has been introduced into the IP header to simplify this classifier). This enables the incoming interfaces downstream of the sender to recognise packets which are part of a reservation, and which should therefore get priority processing in the outgoing interfaces using a

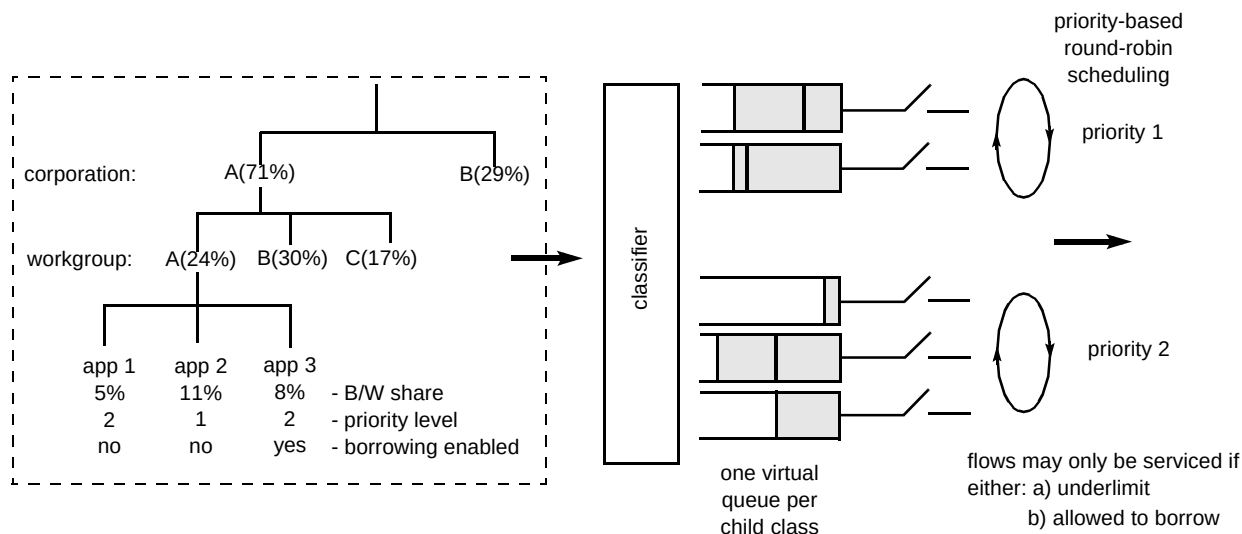


Fig 7 Example of CBQ classification and scheduling

scheduling algorithm such as WFQ or CBQ. The receiver-initiation supports heterogeneity by allowing each receiver to request only the subset of flows it requires, with whatever QoS parameters it decides are necessary. Where requests from multiple receivers converge as they propagate upstream, they are **merged** into a single, 'highest-common denominator' reservation, both giving bandwidth efficiency and protecting the sender from the potential signalling implosion caused by large numbers of *Resv* messages arriving simultaneously. (RSVP is a soft-state protocol, so that regular refreshes of *Path* and *Resv* messages are required.) A number of different reservation **styles** are defined, which determine whether reserved bandwidth is for a specific sender, a specific group of senders, or for any sender within the multicast group.

Currently there are two service classes defined. These are **controlled load** and **guaranteed** service. Controlled load has a loose service definition, which seeks to provide the performance of a 'lightly loaded Internet'. Guaranteed service seeks to provide firm assurances on bandwidth and absolute worst-case end-to-end delay through the use of per-flow scheduling with an algorithm such as WFQ.

RSVP version 1 [1] is now complete. There are a number of known limitations, and restrictions on recommended usage, which have been summarised in a recent IETF statement [21]. Most notably, these relate to scalability. However, rather than address these issues immediately the IETF is proposing to wait until the experience of early deployments has been gathered before proceeding with any further development.

6.7 Differential Services.

Within the IETF it is widely recognised that not only is RSVP unlikely to be deployed widely across the Internet in the foreseeable future, but also that it represents an 'overkill' solution for supporting most of today's applications. Nevertheless, there is a strong consensus that some form of 'differential' quality of service is needed, which would provide a mechanism for certain users to receive a better service, but which could be introduced without the large-scale change necessitated by the widespread deployment of RSVP. Combined with premium charging, such a capability could provide ISPs with an incentive to invest in the infrastructure necessary to improve overall dimensioning and performance.

Discussion is currently taking place on how best to achieve this, and a number of ideas have been proposed [22]. Common to these is the notion of some entity at the access node which regulates a user's access to the network in some way according to a 'user profile'. Beyond the access node, routers have no knowledge of individual user flows, and are only concerned with aggregate traffic, so that the network remains fundamentally connectionless. These schemes may be

seen as a form of end-to-end connectionless bandwidth management, and mechanisms such as WRED, CBQ, etc, described above, are variously proposed as the elements for achieving this. There are many technical and commercial issues that need to be understood, but this is a key area, with the clear potential to provide the Internet's first major step along the route to becoming a true multiservice platform.

7. Asymmetric and unidirectional links

Internet architecture has traditionally assumed networks and routers interconnected via heterogeneous, bi-directional and symmetric links, whether of a point-to-point or broadcast nature. However, the increasing use of unidirectional satellite technology and the planned deployment of broadband asymmetric Internet delivery systems such as ADSL and cable hybrid fibre coaxial (HFC) has led to a reassessment of the Internet protocols and a modification of some of the associated behaviours.

7.1 Asymmetric TCP effects

As demonstrated in Fig 2, TCP data flow relies on a closed loop of data sent and ACKs returned. TCP throughput can easily be limited by the bandwidth of the return channel. While the ACKs may have no data payload, their size is often not insignificant compared with the data packet size (at least 40 bytes TCP/IP headers, in addition to any link-layer framing for the ACKs and commonly a maximum size of 576 bytes including TCP/IP headers for the data packets). This consideration is significant where asymmetric bandwidths are used, especially where the return channel sees more congestion, as in the satellite example below. Two solutions can be used to reduce the return channel bandwidth:

- use link layer compression — standard TCP/IP header compression often reduces the 40 byte TCP/IP header to 1 byte,
- reduce the number of ACKs sent — since TCP uses a cumulative acknowledgement scheme, one ACK can acknowledge multiple datagrams.

Most TCPs employ an ACK-every-other-segment algorithm, and TCP could in principle use an ACK-every-Nth-segment approach (Hughes DirecPC use this approach). There is a down-side to this as it makes the sender traffic pattern more bursty.

7.2 Dynamic IP routing

Internet routing and protocols have been designed to optimise the transmission assuming bi-directional links. Some, but not all, routing protocols **do** take account of asymmetric links (i.e. those bi-directional links with unequal forward and return bandwidths). However, almost

all currently used IP routing protocols fail to support unidirectional links properly. This causes problems in cases such as that shown in Fig 9, where routers A and C are directly connected only by a single unidirectional link, but are indirectly connected via router B, or by an IP network (e.g. where the link between routers A and B may be a dial-up link). A dynamic routing protocol is running between the routers.

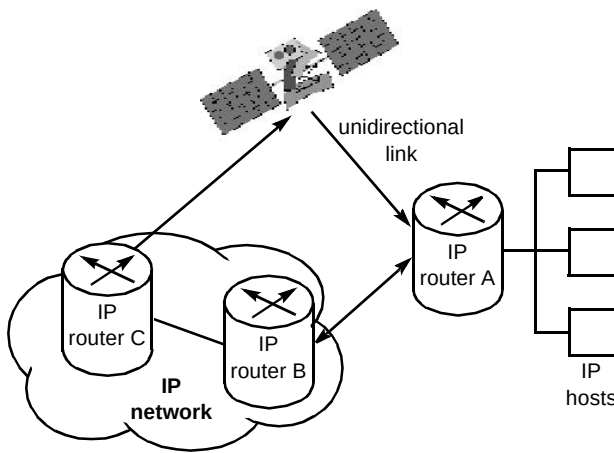


Fig 9 Unidirectional link routing.

The reasons why IP dynamic routing protocols do not work in these cases depend on the particular protocol (e.g. RIP, OSPF, DVMRP). In general, dynamic routing protocols must exchange reachability information between neighbours. The absence of a direct back-channel for the satellite link means that the upstream and downstream routers do not consider each other to be routing neighbours. Modification of the basic routing protocols to accommodate unidirectional links would be eminently possible. This would be particularly simple for link-state protocols such as OSPF, where each router builds a link map of network from flooded link-state advertisements. However, this should be considered a long-term solution still under research. The solution which several teams have pursued is to use IP tunnelling to make the combination of a direct unidirectional link and on half of a bi-directional link appear to the routing protocol as a single direct bi-directional link. For example, with reference to Fig 9, router A forms an IP tunnel to router C, by which routing information can be sent to C by encapsulating it in an IP datagram addressed to C and routed over the IP network. The IP tunnel then appears to the routing protocol as a direct unidirectional link from A to C. Since neighbour reachability information can then be exchanged between A and C in both directions, the satellite link is incorporated in the routing map, without any modification to the operation of the routing protocol. A further difficulty then needs considering. While router C wants to forward all traffic to router A's subnet over the satellite link, router A does not necessarily want to route all its upstream traffic over the IP tunnel via router C. Access to the public Internet might be

more efficiently reached via router B directly. This is relatively straightforward to solve by configuring the tunnel metrics so that the tunnel from A to C is only used for routing information, while the direct link to router B is used for all other traffic. For more details of dynamic routing over unidirectional links, see the IETF Web pages [23].

7.3 Multicast dynamic routing

A further complexity arises over multicast routing. Multicast routing protocols (such as DVMRP, PIM, CBT) are generally based on reverse path forwarding (RPF), or put another way, they use 'come from' routing instead of 'go to' routing. That is, while a unicast protocol selects the shortest path to the destination, multicast routing protocols generally select the shortest path from the source. A problem then arises because most multicast routing protocols, especially those which rely on underlying unicast routing, assume that the shortest path **from** the source is equal to the shortest path **to** the source. In the satellite case, as outlined above, from the perspective of the downstream multicast router, the shortest path to the source could never be over the satellite link since that is unidirectional, hence any multicast packets received over the satellite link would be discarded. There is a solution to this problem, assuming the tunnelling approach to IP routing, which is the direct parallel of the unicast solution, which works provided unicast and multicast use separate routing tables. For multicast routing, the tunnel metrics must be configured so that the shortest path from the downstream multicast router to the head-end router appears to be over the satellite link. This is the exact opposite of the case for unicast routing, where the shortest path from the downstream router to the head-end must be configured to be over the terrestrial link.

8. Internet protocol version 6

The Internet protocol (IP) has its roots in the early academic and latter military networks of 1970s. Only recently, though, its usage started to grow with phenomenal rates and continues to do so as the base protocol for the global Internet network. The Internet Engineering Task Force decided in 1990 that it was time to make some enhancements to this very good, but also very old protocol (considering the developments in computing during the same period). Thus the IPng Working Group was created which after a little less than eight years is at its final stages. The new Internet protocol (IPv6) is about to become a standard opening the way for its widespread use.

IPv6 is a simplified and enhanced version of IPv4 (see Fig 10). The Internet protocol was 'cleared' of fields that were unnecessary (fragmentation, checksum, header options, padding), other field semantics were tidied up (TTL, TOS, length), the header was optimised for hardware implementation (fixed size) and relieved from the problems of the exhausting address space (128 bits from 32-bit

addresses). IPv6 also added a 'class field' and a 'flow label field' for packet and flow-based classification that can be used for QoS provisioning through mechanisms like differential services and the reservation protocol (RSVP).

V4	IHL	type of service	total length	
identification			flags	fragment offset
time to live	protocol		header checksum	
source address				
destination address				
options				padding

V6	class	flow label		
payload length		next header	hop limit	
source address				
destination address				

Fig 10 Comparison of headers, IPv4 (top), IPv6 (bottom).

The IPv6 header is followed by a number of option headers. These are hop-by-hop header, destination header (tunnelling up to a point), routing header, fragment header, authentication header, destination header (final destination), upper-layer header (e.g. TCP, UDP). These headers can be used by a variety of higher layer mechanisms such as source routing, QoS, authentication, tunnelling, mobility, etc, without affecting the forwarding speed of intermediate and uninterested routers.

The addressing structure was significantly changed by introducing hierarchy levels between the backbone sites, the ISPs and the end users. This ensures that aggregation of addresses is achieved which improves routing and forwarding performance. On the other hand, attention has been given to the fact that renumbering (the change of IP address due to, say, change of ISP) could become very frequent and thus very difficult. In IPv6 renumbering is done automatically and it may be possible to provide ISP-independent address aggregates at points of provider/customer interconnect. IPv6 has also embraced two very

important technologies, multicasting and security, that were independent of, and not widely supported in, IPv4. In IPv6 all nodes must have multicast and security (IPSEC) capabilities which opens the way for a variety of applications from multicast gaming to secure electronic commerce. Finally, neighbour discovery (ND), stateful and stateless address autoconfiguration provides maximum configuration flexibility for any use of the Internet protocol — from no to minimum configuration required for isolated (e.g. home) LANs, to full control over every configuration parameter for an Internet service provider (ISP) or a corporate intranet [24].

The main problem for IPv6 today is that it does not have any 'killer application' because CIDR (classless inter-domain routing) and NAT (network address translator) have allayed fears over the exhaustion of the address space while many of the IPv6 features have been retrofitted into IPv4. IPv6 is an altogether better protocol than IPv4 but there is no **one** reason why someone would move to IPv6. However, the demand for IP header priority bits and the flow_id for quality-of-service support might develop into one. Even with such a reason, interoperation with IPv4, and hence evolutionary deployment, is still a big problem although solutions are now starting to emerge through the development of protocol translators (PTs) to application layer gateways (ALGs). Most of the initial design work on IPv6 has now been completed and vendor, market and experimental feedback will decide its fate. It is likely, however, that many aspects of IPv6 will continue to be developed during the next few years. It is also almost certain that it will take years (if ever) before IPv6 completely replaces its predecessor (IPv4) and much effort will need to be invested on the migration and interoperation mechanisms for the two versions of the Internet protocol.

9. High-speed routing

As previously stated, a key assumption for the future delivery of differential services and, ultimately, end-to-end QoS, is that the underlying network will be adequately dimensioned. The strain on the current Internet backbone is already apparent. Core routers must hold approximately 45 000 routes in a stable fashion for current connection levels. Routing table look-up and subsequent forwarding under the increasingly heavy traffic loads have become the major performance bottle-neck. All proposed future services will require routers to hold even more 'look-up' state as well as the ability to perform possibly complex traffic scheduling algorithms on a per-packet basis. The need for more scalable router architectures has become very important and especially so for any nodes forming the interconnection points between large carrier backbones and the designated IXP (Internet exchange point) sites within the global Internet topology.

One thing that has led the recent evolution of routing and forwarding in the Internet has been the assumption that routing look-ups are not fast enough for software-based

gigabit switching. This is because routeing tables are too big to fit in an on-chip cache, while an off-chip cache is too expensive or too slow for gigabit routeing. From this assumption a lot of technologies have been introduced to the Internet (e.g. ATM, IP switching, tag switching, etc). The target of all these technologies is to bypass IP layer forwarding and do it instead in layer 2.

Recent studies, however, are challenging the above basic assumption and in effect layer 2 switching all together, as a means of making routeing faster. In Degermark et al [25] the authors argue that you can do gigabit forwarding on PentiumPro and Alpha21164 processors. They do that by minimising the number of memory accesses required for a look-up, while they also minimise the size of the data structure (routeing entries in the form of a tree). Using clever algorithms they manage to fit large routeing tables (~40 000 entries) in 150—160 Kbytes which goes in the secondary cache on-chip memory of a PentiumPro. Combining this with the minimisation of memory access it is predicted that up to two million routeing look-ups per second may be achieved. With a packet size of 1000 bits (125 bytes) this is equivalent to 2 Gbit/s throughput on your home computer!

10. Conclusions

The Internet is a collection of domains structured into routeing, addressing and business hierarchies for the global delivery of packet based multimedia applications. It is a very dynamic networking, commercial and political environment where change is the only constant. In this paper, the Internet architecture has been described in terms of its underlying design concepts and protocol components. These have focused on the major elements at a point in their rapid evolution and should not be considered to be definitive or complete. The latest status of the Internet protocols and many of the well-known applications can be determined from the documents of the Internet Engineering Task Force at <http://www.ietf.org/>. The views expressed in this paper are those of the authors and should not be considered to represent either a BT or an IETF consensus view.

References

- 1 Braden R et al: 'Resource ReSerVation Protocol (RSVP) — Version 1 Functional Specification', RFC 2205, (September 1997), <ftp://ds.internic.net/rfc/rfc2205.txt>
- 2 Postel J: 'Transmission control protocol', RFC793, (September 1981), <ftp://ds.internic.net/rfc/rfc793.txt>
- 3 Stevens W: 'TCP slow start, congestion avoidance, fast retransmit, and fast recovery algorithms', RFC 2001, (January 1997), <ftp://ds.internic.net/rfc/rfc2001.txt>
- 4 Braden R et al: 'Integrated Services in the Internet Architecture: an Overview', RFC 1633, (June 1994), <ftp://ds.internic.net/rfc/rfc1633.txt>
- 5 Newson D J, Ginsburg D and Williams M T: 'Next generation local area networks', BT Technol J, **16**, No 1, pp 76—87 (January 1998).
- 6 Plummer D C: 'Ethernet Address Resolution Protocol', RFC 826, (November 1982), <ftp://ds.internic.net/rfc/rfc826.txt>

- 7 Braden R: 'Requirements for Internet hosts — communication layers', RFC1122, (October 1989), <ftp://ds.internic.net/rfc/rfc1122.txt>
- 8 Armitage G: 'Support for Multicast over UNI 3.0/3.1 based ATM Networks', RFC 2022, (November 1996), <ftp://ds.internic.net/rfc/rfc2022.txt>
- 9 Bormann C: 'Providing integrated services over low-bitrate links', (May 1997), <ftp://ftp.ietf.org/internet-drafts/draft-ietf-issll-isslow-02.txt>
- 10 Simpson W: 'The Point-to-Point Protocol (PPP)', RFC1661, (July 1994), <ftp://ds.internic.net/rfc/rfc1661.txt>
- 11 Everett C H W, Blakey K M and Morgan L N C: 'The role of ISDN in data networking', BT Technol J, **16**, No 1, pp 39—51 (January 1998).
- 12 Sklower M K et al: 'The PPP Multilink Protocol (MP)', RFC1990, (August 1996), <ftp://ds.internic.net/rfc/rfc1990.txt>
- 13 Richards C and Smith K: 'The PPP Bandwidth Allocation Protocol (BAP). The PPP Bandwidth Allocation Control Protocol (BACP)', RFC2125, (March 1997), <ftp://ds.internic.net/rfc/rfc2125.txt>
- 14 Hamzeh K et al: 'Layer Two Tunneling Protocol 'L2TP'', (August 1997), <ftp://ftp.ietf.org/internet-drafts/draft-ietf-pppext-l2tp-06.txt>
- 15 The IP Multicast Initiative WWW pages, <http://www.ipmulticast.com/>
- 16 Whalley G M et al: 'Advanced internetwork design — today and in the future', BT Technol J, **16**, No 1, pp 25—38 (January 1998).
- 17 O'Neill A: 'Internetwork futures', BT Technol J, **15** No 2, pp 226—240 (April 1997).
- 18 Clark D et al: 'Supporting real-time applications in an integrated services packet network: architecture and mechanisms', Proc Sigcom'92 (1992).
- 19 Wakeman I et al: 'Implementing real time packet forwarding policies using streams', Usenix 1995 Technical conference, New Orleans, pp 71—82 (1995), <ftp://cs.ucl.ac.uk/darpa/usenix-cbq.ps>
- 20 White P: 'RSVP and Integrated Services in the Internet: A Tutorial', IEEE Communications Magazine (May 1997).
- 21 Mankin A et al: 'Resource reservation protocol (RSVP), Version 1, Applicability statement: some guidelines on deployment', RFC 2208 (September 1997), <ftp://ds.internic.net/rfc/rfc2208.txt>
- 22 Clark D and Wroclawski J: 'An approach to service allocation in the Internet', (July 1997), <ftp://ftp.ietf.org/internet-drafts/draft-clark-diff-svc-alloc-00.txt>
- 23 IETF Unidirectional Link Routing pages, <http://www.ietf.org/html.charters/udlr-charter.html>
- 24 Deering S and Hinden R: 'Internet protocol, version 6 (IPv6) specification', (July 1997), <ftp://ftp.ietf.org/internet-drafts/draft-ietf-ipngwg-ipv6-spec-v2-00.txt>
- 25 Degermark M et al: 'Small Forwarding Tables for Fast Routing Lookups', SIGCOMM '97 (1997).



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Martin Tatham studied at Christchurch, Oxford University, and was awarded a first in Physics followed by a DPhil for his thesis on the study of sub-picosecond electron-phonon dynamics in semiconductor quantum wells. He joined BT Laboratories in 1989. His initial work at BT was in the field of semiconductor optoelectronics, involving the modelling of long wavelength semiconductor quantum well lasers. This was followed by studies of high-speed lasers and of nonlinear gain in lasers. In 1994 he joined the Networks Modelling group, to work on the modelling of future high capacity optical networks, and on

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Simon Carter obtained a BSc in Physics and a PhD for research in Lock-in Amplifiers from the University of Reading. After working first for Hewlett-Packard, and then for York Technology, he joined BT Laboratories (BTL) in 1987. For the first few years he worked on undersea optical systems. Later, he joined the Networks Research Centre at BTL, and worked on a number of projects related to the design of optical networks. Two years ago his interests transferred to the future of IP networking, with the support of quality of service as a particular interest.



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